Profiling

Modelització Estadística

Grau en Intel·ligència Artificial

Course 2024-25

Outline

Introduction to Profiling

What is Profiling?

Profiling is the process of describing and interpreting the characteristics of clusters identified from clustering algorithms.

- **Objective**: Identify significant variables that differentiate clusters.
- **Focus**: Summarize the structure of clusters using both:
  - Continuous variables (e.g., mean, median).
  - Categorical variables (e.g., proportions, distributions).

Applications of Profiling

Profiling clusters is widely used in various fields:

- **Marketing**: Segment customers based on behavior or demographics.
- **Healthcare**: Identify patient subgroups for personalized treatments.
- **Sociology**: Analyze patterns in social data (e.g., communities, opinions).
- **Business Intelligence**: Understand groups in sales or operational data.

Why is Profiling Important?

- **Interpretability**: Profiling explains the structure and characteristics of clusters.
- **Actionability**: Provides insights to tailor actions based on cluster traits.
- **Validation**: Helps confirm that the clusters are meaningful and distinguishable.

Descriptive Analysis for Continuous Variables. Statistics

Summary Statistics for Continuous Variables

To describe continuous variables for each cluster, use key summary statistics:

- **Mean** and **Standard Deviation**: Indicate central tendency and variability.
- **Median** and **Interquartile Range (IQR)**: Robust measures for non-normal distributions for central tendency and variability.
- **Minimum and Maximum**: Highlight the range of values within each cluster.

Example

Computing Summary Statistics in R:

- Use aggregate() to summarize variables across clusters.

```
set.seed(12345)
kmeans_result <- kmeans(mtcars,3)
mtcars$clust <- factor(kmeans_result$cluster)
aggregate(mtcars$mpg,
          by  = list(cluster = mtcars$clust),
          FUN = summary)
```

```
##   cluster   x.Min. x.1st Qu. x.Median   x.Mean x.3rd Qu.   x.Max.
## 1       1 15.20000  15.35000 16.40000 17.01429  17.70000 21.40000
## 2       2 17.80000  21.00000 22.80000 24.50000  28.07500 33.90000
## 3       3 10.40000  13.30000 14.70000 14.64444  15.80000 19.20000
```

Descriptive Analysis for Continuous Variables. Graphics

Visualizing Continuous Variables

Visualizations help interpret differences between clusters:

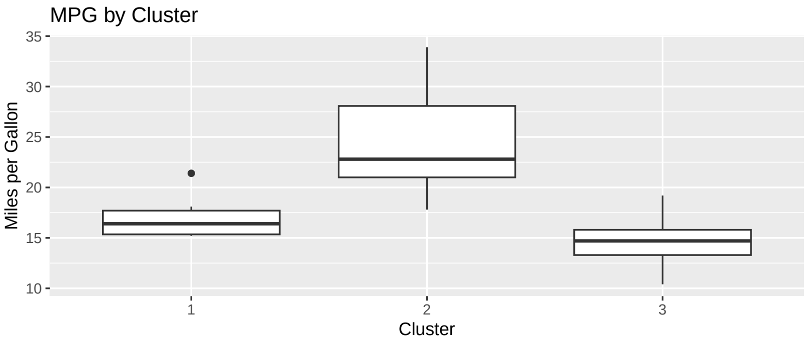
- **Boxplots**: Compare the distribution of a variable across clusters.
- **Histograms**: Highlight differences in distributions.
- **Density Plots**: Highlight differences in distributions.
- **Violin Plots**: Visualize density and spread.

Example

Creating **Boxplots** in R:

- Use `geom_boxplot` in combination with `ggplot`.

```
library(ggplot2)
# Boxplot of MPG by Cluster
ggplot(mtcars, aes(x = clust, y = mpg)) +
  geom_boxplot() +
  labs(title="MPG by Cluster",x="Cluster",y="Miles per Gallon")
```

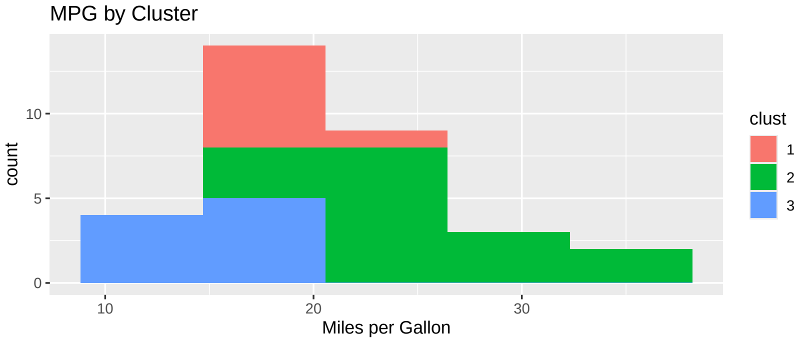


Example

Creating **histograms** in R:

- Use `geom_histogram` in combination with `ggplot`.

```r
# Histogram of MPG by Cluster
ggplot(mtcars, aes(x = mpg, fill = clust)) +
  geom_histogram(bins = 5) +
  labs(title = "MPG by Cluster", x = "Miles per Gallon")
```

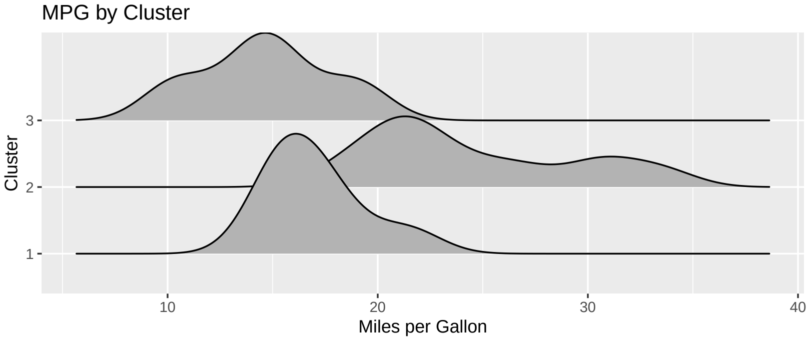


Example

Creating **density plots** in R:

- Use `geom_density_ridges` in combination with `ggplot`.

```
library(ggridges)
# Density of MPG by Cluster
ggplot(mtcars, aes(x = mpg, y = clust, group = clust)) +
  geom_density_ridges() +
  labs(title="MPG by Cluster",x="Miles per Gallon",y="Cluster")
```

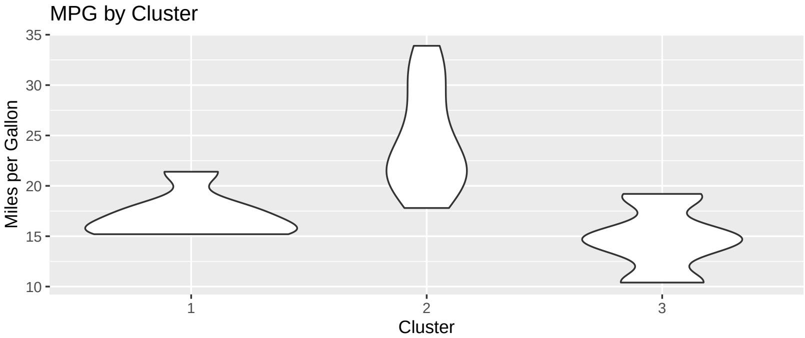


Example

Creating **Violinplots** in R:

- Use `geom_violin` in combination with `ggplot`.

```
# Violinplot of MPG by Cluster
ggplot(mtcars, aes(x = clust, y = mpg)) +
  geom_violin() +
  labs(title="MPG by Cluster",x="Cluster",y="Miles per Gallon")
```



Graphical options for profiling continuous variables

- **Boxplots**:
  - Summarize distributions using median, quartiles, and outliers.
  - Best for comparing central tendencies and spread across clusters.
- **Histograms**:
  - Show frequency distribution within clusters.
  - Ideal for identifying skewness and bin-based patterns.
- **Density Plots**:
  - Provide a smoothed representation of data distribution.
  - Great for visualizing overlapping distributions and detecting modes.
- **Violin Plots**:
  - Combine boxplot summary with density visualization.
  - Useful for comparing the distribution shape and spread in detail.

Descriptive Analysis for Categorical Variables. Statistics

Summary for Categorical Variables

To describe categorical variables within clusters:

- **Proportions**: Compute the proportion of each category within each cluster.
- **Contingency Tables**: Summarize the relationship between clusters and categories.
- **Counts**: Highlight absolute frequencies for context.

Example

- Use `table()` to create **contingency tables**.
- Use `prop.table()` to create **proportion tables**.

```
# Contingency table: Gear by cluster
with(mtcars, table(clust, gear))
```

```
##      gear
## clust  3  4  5
##     1  7  0  0
##     2  1 12  3
##     3  7  0  2
```

```
# Proportion table: Gear by cluster
round(with(mtcars, prop.table(table(clust, gear),1)),2)
```

```
##      gear
## clust    3    4    5
##     1 1.00 0.00 0.00
##     2 0.06 0.75 0.19
##     3 0.78 0.00 0.22
```

Descriptive Analysis for Categorical Variables. Graphics

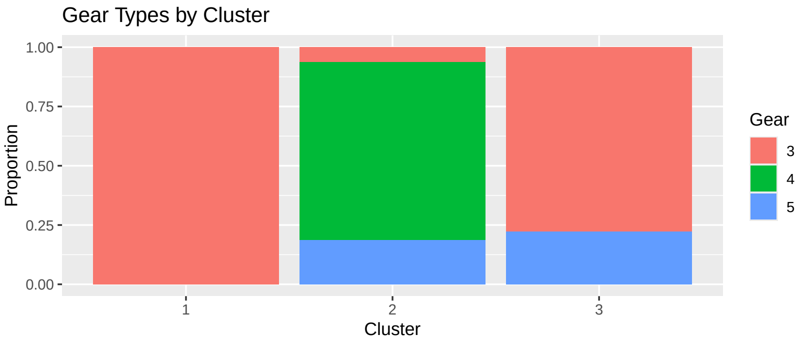
Visualizing Categorical Variables

- **Stacked Bar Charts**: Compare category proportions across clusters.
- **Mosaic Plots**: Highlight relationships between clusters and categories.

Example

- Use `geom_bar` in combination with `ggplot`.
- Use `position = "fill"` for proportional bar charts.

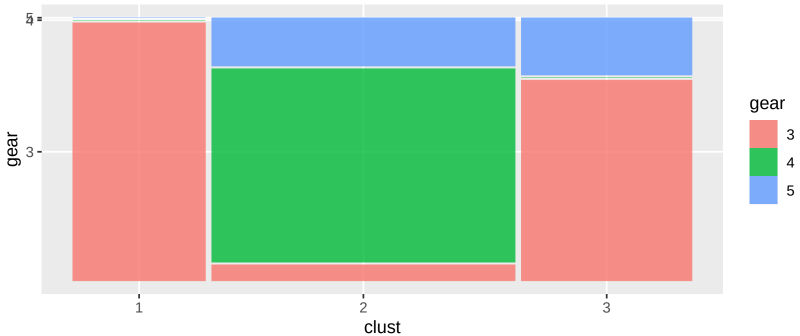
```r
# Bar plot of Gear by Cluster
ggplot(mtcars, aes(x = clust, fill = factor(gear))) +
  geom_bar(position = "fill") +
  labs(title="Gear Types by Cluster",x="Cluster",y="Proportion") +
  scale_fill_discrete(name = "Gear")
```



Example

- Use `geom_bar` in combination with `ggplot`.
- Use `position = "fill"` for proportional bar charts.

```r
library(ggmosaic)
# Mosaic plot of Gear by Cluster
mtcars$gear  <- factor(mtcars$gear)
mtcars$clust <- factor(mtcars$clust)
ggplot(data = mtcars) +
  geom_mosaic(aes(x = product(clust), fill = gear))
```



Radar Plots

What are Radar Plots?

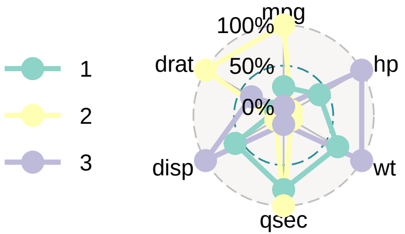
Radar plots (or spider plots) visually represent multivariate data by displaying variables as axes arranged in a circular layout:

- Each cluster is plotted as a polygon, connecting points that represent scaled variable values.
- **Purpose**: Compare clusters across multiple variables simultaneously.
- Highlights **differences** in cluster profiles by visualizing their "shape."

Example

- Creating Radar Plots in R with ggplot2 and ggradar.
- Variables are normalized to ensure comparability.

```r
# devtools::install_github("ricardo-bion/ggradar")
library(ggradar); library(dplyr); library(scales); library(tibble)
normalize <- function(x, na.rm = TRUE) (x- min(x))/(max(x)-min(x))
# Summarize cluster profiles
cluster_summary <- mtcars %>% group_by(clust) %>%
  summarise(across(c(mpg, hp, wt, qsec, disp, drat), mean)) %>%
  mutate(across(-clust, normalize))  %>% select(-clust) %>%
  as_tibble(rownames = "group")
ggradar(cluster_summary)
```



Why Use Radar Plots for Clusters?

- **Multivariate Comparison**: Radar plots summarize multiple variables in a single, intuitive visualization.
- **Cluster Profiling**: Quickly identify which variables differentiate clusters.
- **Limitations**: Best used with a moderate number of variables (5-10) to avoid excessive overlapping.

K-prototypes

Example

The function `clprofiles` of the package `clustMixType` allows to build several graphics for comparing variables between clusters after using K-Prototypes.

```
library(clustMixType)
kp <- kproto(mtcars[,1:11], k=3, type = "gower")
clprofiles(kp, mtcars)
```

Introduction to the `DataExplorer` Package

What is `DataExplorer`?

The `DataExplorer` package simplifies data exploration and profiling with automated visualizations:

- Useful for quickly **summarizing** data distributions, relationships, and missing values.
- **Cluster Analysis**: Apply the package to visualize how clusters differ across multiple variables.

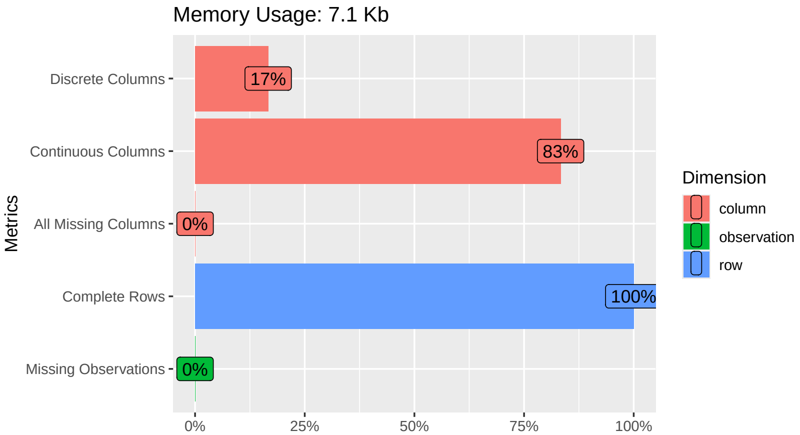
Key Functions for Cluster Profiling

- `plot_intro()`: Summarizes the dataset (e.g., variable types, missing values).
- `plot_boxplot()`: Visualizes distributions of continuous variables grouped by clusters.
- `plot_bar()`: Displays categorical variable distributions across clusters.
- `plot_correlation()`: Examines correlations between variables.

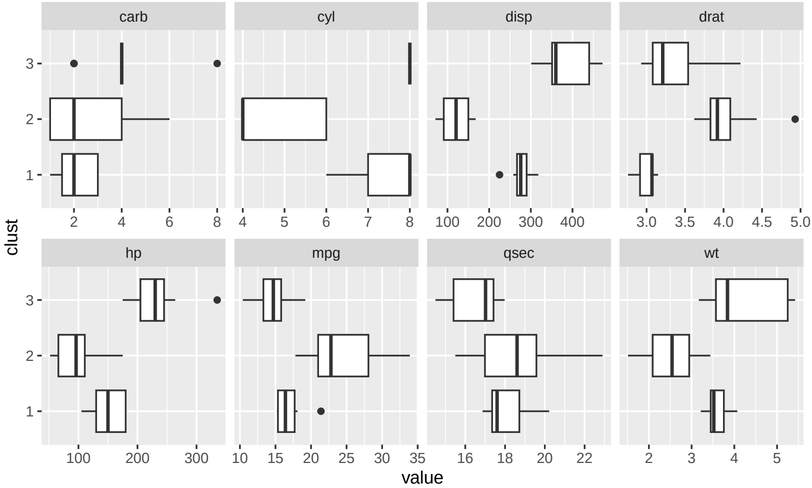
Example

Combine plots to gain insights into multivariate cluster profiles.

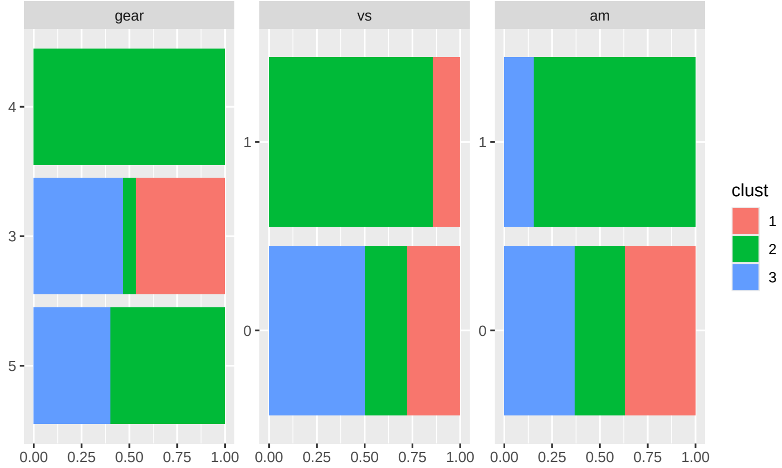
```r
# Load DataExplorer
library(DataExplorer)
# Summarize data
plot_intro(mtcars)
```



```r
# Boxplot of continuous variables by cluster
plot_boxplot(mtcars, by = "clust")
```



```r
# Bar plots for categorical variables (e.g., gear by cluster)
plot_bar(mtcars, by = "clust")
```



```r
# Correlation plot Cluster 1
plot_correlation(mtcars[mtcars$clust==1,1:11], type="continuous")
```

Tests. Introduction

Outline

We will see the following tests:

- ANOVA test
- Kruskal-Wallis test
- Chi-square test

ANOVA. Statistical Tests

What is ANOVA?

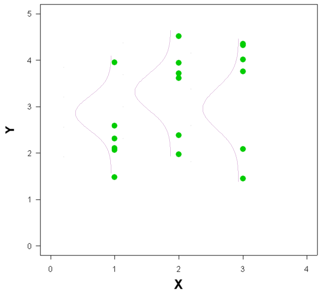
Analysis of Variance (**ANOVA**) compares the means of a continuous variable across multiple groups (clusters):

- Determines if there is a **statistically significant difference** between group means.
- Assumes:
  - **Normally distributed data** within each group.
  - **Homoscedasticity**. Homogeneity of variances across groups.

ANOVA. Rationale

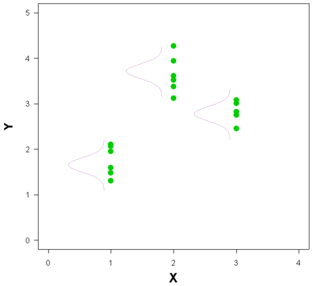
- The following magnitudes are compared:
  - The **variance of all the data** regardless the cluster
  - The average of the **within-cluster** variances
- If they are similar, there are no mean differences and vice versa.

No mean differences



Intra-group variability similar to total variability

Mean differences



Intra-group variability different to total variability

ANOVA. Hypothesis Testing

Hypothesis Testing in ANOVA

To test whether there is a significant difference between the means of k groups (clusters), the following hypotheses are formulated:

- **Null Hypothesis (H_0)**: All group means are equal.

$$H_0 : \mu_1 = \mu_2 = \cdots = \mu_k$$

- **Alternative Hypothesis (H_1)**: At least one group mean is different.

$$H_a : \exists\, i, j \text{ such that } \mu_i \neq \mu_j$$

ANOVA. Steps

Steps in ANOVA

1. Calculate the **Between-Group Variance**:

$$SSB = \sum_{i=1}^{k} n_i(\bar{x}_i - \bar{x})^2$$

where:
- n_i: Number of observations in cluster i.
- $\bar{x}_i$: Mean of cluster i.
- $\bar{x}$: Overall mean.

2. Calculate the **Within-Group Variance**:

$$SSW = \sum_{i=1}^{k} \sum_{j=1}^{n_i} (x_{ij} - \bar{x}_i)^2$$

3. Compute the **F-Statistic**:

$$F = \frac{MSB}{MSW} = \frac{SSB/(k-1)}{SSW/(n-k)}$$

Compare F to a critical value from the $F(k\text{-}1, n\text{-}k)$.

ANOVA. Table

Structure of the ANOVA Table

The ANOVA table breaks down the variance into components to test the hypothesis $H_0 : \mu_1 = \mu_2 = \cdots = \mu_k$:

Source of Variation	SS	df	MS	F	p
Between Variance	SSB	$k-1$	$\text{MSB} = \frac{\text{SSB}}{k-1}$	$\frac{\text{MSB}}{\text{MSW}}$	p
Within Variance	SSW	$n-k$	$\text{MSW} = \frac{\text{SSW}}{n-k}$	-	-
Total	SST	$n-1$	-	-	-

- *SS*: Sum of squares
- *df*: Degrees of Freedom
- *MS*: Mean Square
- *F*: F-Statistic
- *p*: p-value

ANOVA. Example

Example

A **small p-value ($p < 0.05$)** suggests significant mean differences by clusters.

- Regarding the mpg, there are mean differences between clusters
- Regarding the qsec, there are also mean differences between clusters

```
anova_result <- aov(mpg ~ clust, data = mtcars)
summary(anova_result)
```

```
##             Df Sum Sq Mean Sq F value  Pr(>F)
## clust        2  644.3   322.1   19.39 4.5e-06 ***
## Residuals   29  481.8    16.6
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary(aov(qsec  ~ clust, data = mtcars))
```

```
##             Df Sum Sq Mean Sq F value  Pr(>F)
## clust        2  27.00  13.499   5.438 0.00987 **
## Residuals   29  71.99   2.482
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

ANOVA. Interpretation and assumptions

Interpreting ANOVA Results

- **Significant Result**: If $p < 0.05$, at least one group mean differs from the others.
- **Post-Hoc Tests**: Use pairwise comparisons (e.g., Tukey's HSD) to identify which groups differ.
  - `TukeyHSD()`
- **Check Assumptions**:
  - Normality
    - Numerically: `shapiro.test()`.
    - Graphically: `qqnorm()`.
  - Homoscedaticity (Homogeneity of variances)
    - Numerically: `bartlett.test()` or `leveneTest()`.
    - Graphically: `boxplot()`.

ANOVA. Tukey test

Multiple comparison

Tukey test performs pairwise cluster comparisons correcting p-values by multiplicity.

```
TukeyHSD(anova_result)
```

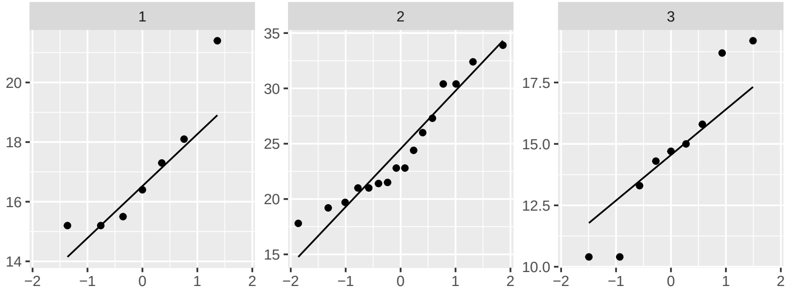
```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = mpg ~ clust, data = mtcars)
##
## $clust
##           diff        lwr       upr     p adj
## 2-1  7.485714   2.924178 12.047250 0.0009821
## 3-1 -2.369841  -7.442620  2.702937 0.4897051
## 3-2 -9.855556 -14.049715 -5.661396 0.0000080
```

ANOVA. Normality assumption

```
# p-values of shapiro.test
with(mtcars,unlist(lapply(tapply(mpg, clust, shapiro.test),
                                  '[[','p.value')))
```

```
##          1          2          3
## 0.09734829 0.14859508 0.46539935
```

```
## QQ-norms
ggplot(mtcars,aes(sample = mpg)) + labs(x="", y="") +
  geom_qq() + geom_qq_line() + facet_wrap(~clust,scales="free_y")
```

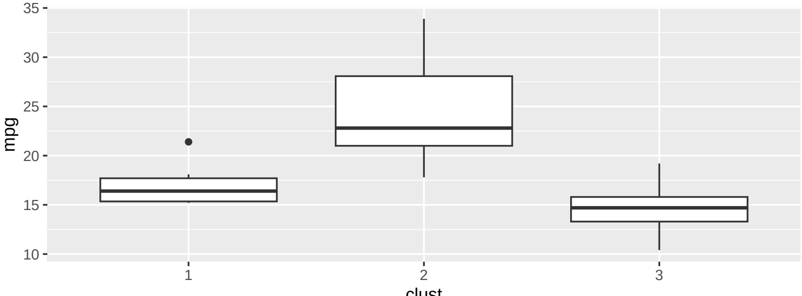


ANOVA. Homoscedasticity assumption

```r
## Bartlett's test
with(mtcars, bartlett.test(mpg ~ clust))
```

```
##
##  Bartlett test of homogeneity of variances
##
## data:  mpg by clust
## Bartlett's K-squared = 5.2318, df = 2, p-value = 0.0731
```

```r
## Boxplots
ggplot(mtcars,aes(x = clust, y = mpg)) +
  geom_boxplot()
```



Assumptions

Assumptions

Both assumptions (Normality and Homoscedasticity) are **met**.

Kruskal-Wallis. Statistical Tests

What is the Kruskal-Wallis Test?

The **Kruskal-Wallis Test** is a non-parametric alternative to ANOVA for comparing continuous variables across groups:

- **Objective**: Test if the distributions of a continuous variable differ significantly across clusters.
- Does **not assume**:
  - Normality of data.
  - Homogeneity of variances.
- Based on **ranks** rather than raw data.

Kruskal-Wallis Test. Steps

Steps in the Kruskal-Wallis Test

1. Rank all data points from smallest to largest, across all groups.
2. Compute the **test statistic**:

$$H = \frac{12}{n(n+1)} \sum_{i=1}^{k} n_i R_i^2 - 3(n+1)$$

   where:

   - n_i: Number of observations in group i.
   - R_i: Sum of ranks for group i.
   - n: Total number of observations.

3. Compare H to a critical value from the chi-square distribution with $k-1$ degrees of freedom.

Kruskal-Wallis Test. Example

Example

Variable drat does not hold **homoscedasticity**.

- According to Kruskal-Wallis test, there are mean differences between clusters in drat.

```
# p-values of shapiro.test denotes that Normality is not met by "drat"
with(mtcars,unlist(lapply(tapply(drat, clust, shapiro.test),
                          '[[','p.value')))
```

```
##           1           2           3
## 0.007602034 0.014251096 0.113845261
```

```
# Comparing RatiFin across clusters
kruskal_result <- kruskal.test(drat ~ clust, data = mtcars)
kruskal_result
```

```
##
##  Kruskal-Wallis rank sum test
##
## data:  drat by clust
## Kruskal-Wallis chi-squared = 19.77, df = 2, p-value = 5.092e-05
```

Kruskal-Wallis Test. Example

Interpreting Kruskal-Wallis Results

- **Significant Result**: If $p < 0.05$, at least one group distribution differs.
- **Post-Hoc Tests**: Use pairwise comparisons (e.g., Dunn's Test) to identify specific group differences: `dunn.test()` or `DunnTest()`
- **Robustness**: Suitable for data with outliers or non-normal distributions.

```r
library(FSA)
dunnTest(mpg ~ clust, data = mtcars)
```

```
##   Comparison         Z       P.unadj        P.adj
## 1      1 - 2 -2.860373 4.231433e-03 8.462866e-03
## 2      1 - 3  1.024724 3.054933e-01 3.054933e-01
## 3      2 - 3  4.350310 1.359455e-05 4.078365e-05
```

Chi-Square Test. Statistical Tests

What is the Chi-Square Test?

The **Chi-Square Test** (χ^2) assesses the association between two categorical variables:

- **Objective**: Test if the distribution of a categorical variable differs significantly across clusters.
- Assumes:
  - Observations are independent.
  - Expected frequencies in each cell are ≥ 5 for validity.

Chi-Square Test. Steps

Steps in the Chi-Square Test

1. Create a **contingency table** of observed frequencies:

$$\text{Table}: \quad \text{Category} \times \text{Cluster}$$

2. Compute the **expected counts** under the null hypothesis:

$$E_{ij} = \frac{R_i \cdot C_j}{n}$$

   where:

   - R_i: Row total for category i.
   - C_j: Column total for cluster j.
   - n: Total number of observations.

3. Calculate the **test statistic**:

$$\chi^2 = \sum_{i=1}^{r} \sum_{j=1}^{c} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

   where O_{ij} and E_{ij} are the observed and expected counts.

4. Compare χ^2 to the critical value from the chi-square distribution with $(r-1)(c-1)$ degrees of freedom.

Chi-Square Test. Example

Example

- The variable gear (as categorical) is significantly different ($p < 0.05$) between clusters.

```
# Testing the association between gear type and clusters
chi_result <- with(mtcars,chisq.test(table(gear, clust)))
chi_result
```

```
##
##  Pearson's Chi-squared test
##
## data:  table(gear, clust)
## X-squared = 25.126, df = 4, p-value = 4.746e-05
```

Chi-Square Test. Interpretation

Interpreting Chi-Square Results

- **Significant Result**: If $p < 0.05$, the distribution of the categorical variable differs significantly across clusters.
- **Check Assumptions**:
  - Ensure all expected counts are ≥ 5. If not, use Fisher's Exact Test.
  - Use `chisq.test()` for standard tests and `simulate.p.value = TRUE` for small samples.

PCA for Comparing Clusters

Why Use PCA for Clusters?

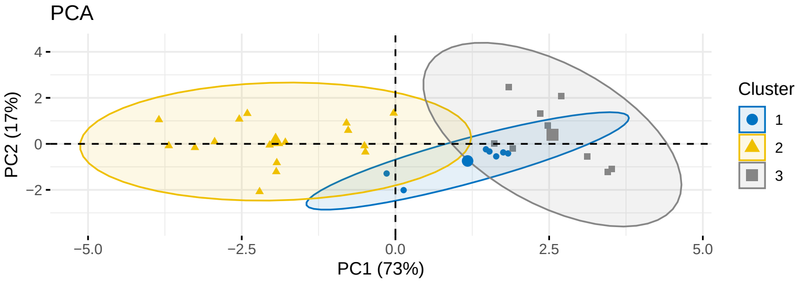
Principal Component Analysis (**PCA**) reduces the dimensionality of multivariate data while preserving variance:

- **Objective**: Visualize clusters in a low-dimensional space (e.g., 2D or 3D).
- Highlights differences between clusters based on the most informative combinations of variables (principal components).
- Allows assessment of cluster separation and overlap in multivariate space.

Example

- Look for distinct groupings of clusters along the principal components.
- Overlap indicates potential cluster similarity or poor separation.

```
library(ggplot2); library(factoextra)
pca_result <- prcomp(mtcars[, 1:7], scale = TRUE) # numeric vars
# Visualize PCA with clusters
fviz_pca_ind(pca_result, geom = "point",
             col.ind=factor(mtcars$clust), palette="jco",
             addEllipses=TRUE, legend.title = "Cluster") +
  labs(title = "PCA", x = "PC1 (73%)", y = "PC2 (17%)")
```



CART to Rank Variable Importance for Clusters

Why Use CART for Cluster Profiling?

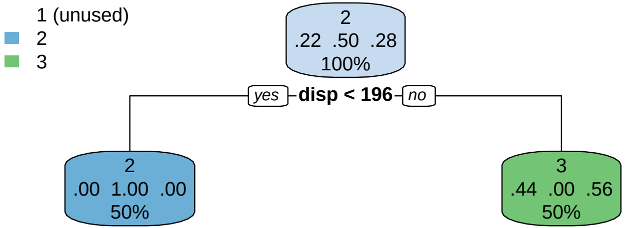
Classification and Regression Trees (**CART**) can rank variable importance based on their ability to separate clusters:

- **Outcome**: Clusters are treated as the dependent (target) variable.
- **Predictors**: All other variables are used to predict cluster membership.
- **Interpretation**: Variables at the top of the tree are the most important for distinguishing clusters.

Example

- fit$variable.importance ranks variables based on the cluster contribution
- Variables higher in the tree are important for defining clusters.

```
library(rpart); library(rpart.plot)
fit <- rpart(clust ~ ., mtcars,method = "class"); rpart.plot(fit)
```




```
fit$variable.importance # Variable importance ranking
```

```
##      disp       cyl      drat        hp       mpg        wt
## 12.062500 10.554688 10.554688  9.800781  9.800781  9.800781
```

Key Takeaways

- **Interpret Top Nodes**: Variables at the top splits are the most critical for cluster discrimination.
- **Use Variable Importance**: The `variable.importance` metric provides a quantitative ranking.
- **Validate Results**: Cross-validate the tree to confirm the robustness of the variable importance rankings.

References

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2. Han, J., Kamber, M., & Pei, J. (2011). Data mining: Concepts and techniques (3rd ed.). Morgan Kaufmann. https://doi.org/10.1016/C2009-0-61819-5